Chapter-7

Data link Control and Protocol

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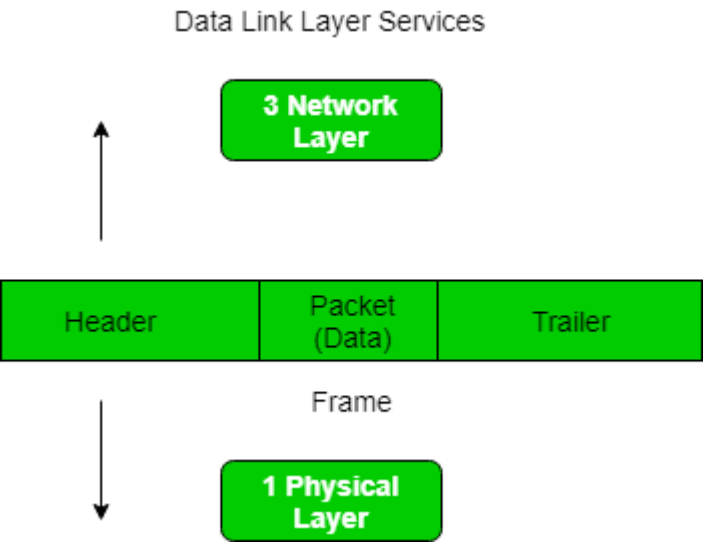
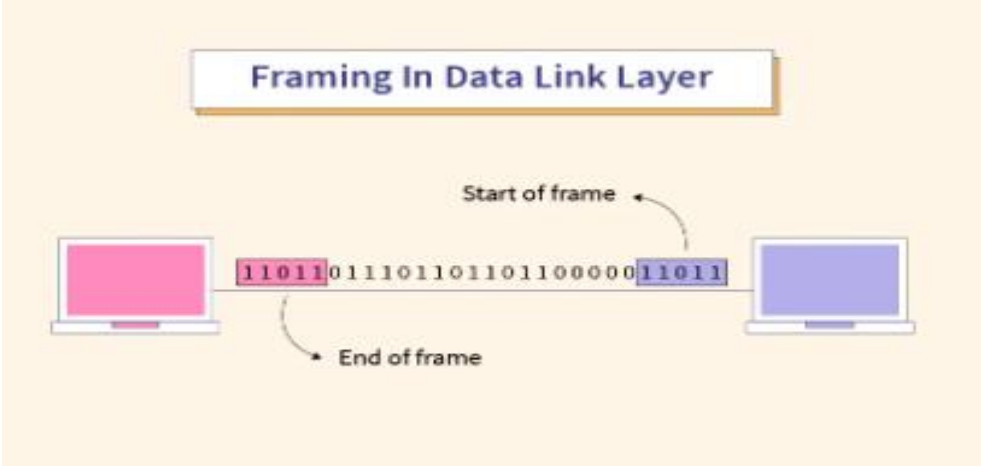
Framing

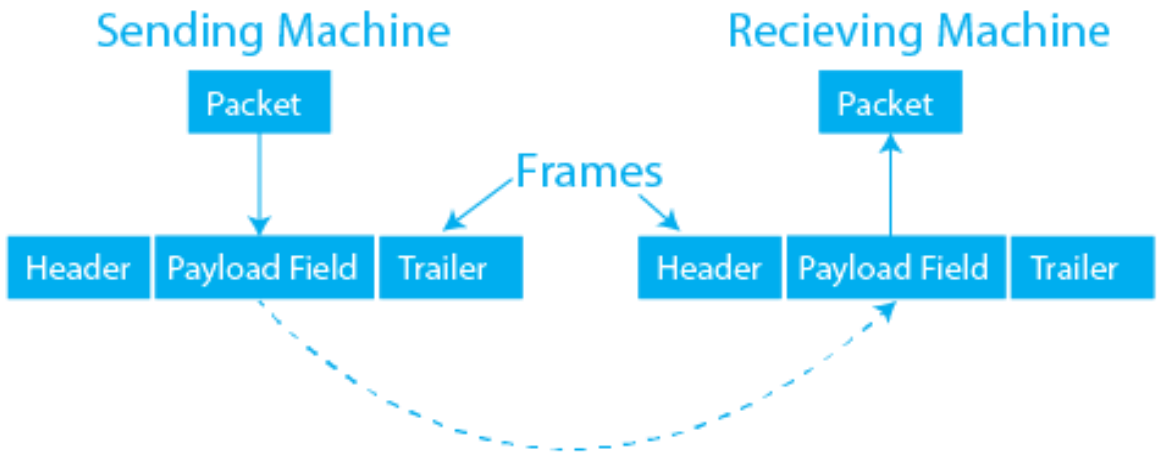
- The data is transferred from one device to another user using various technologies and protocols.
- The digital data transmission unit in communications and computer networking is known as a frame. A frame is a simple network message in systems that use packet switching.
- Framing is a point-to-point connection between two computers or devices consisting of a wire in which data is transmitted as a stream of bits or a series of symbols.





- **Parts of Frame**
- A frame generally has the following parts:
- **Frame Header:** It contains the source and destination address of the frame.
- **Payload Field:** It will contain the message which is needed to be delivered.
- **Trailer:** It contains the error correction and error detection bits.
- **Flag:** It marks the beginning and the end of the frame.

- Framing is a process of dividing a stream of data into smaller, more manageable units called frames.
- The frames are then transmitted over the network and reassembled at the receiving end to recreate the original data stream. Framing is important in computer networks.
- Framing is usually dealt with at the program level by a network protocol like the **Point-to-Point Protocol (PPP)** or the **High-Level Data Link Control (HDLC)**.
- The source and target identifiers, error-checking codes, and other control information are added as headers and trailers by these protocols to each frame.



Types of Framing in Computer Networks

- There are mainly two types of framing
- **Fixed Size Framing:**

The frame is of fixed size and there is no need to provide boundaries to the frame, the length of the frame itself acts as a delimiter.

As a result, it is not necessary to use additional boundary bits to specify the beginning and end of the frame. ATM cells are an example of this.

- **Variable size Framing:** In this, there is a need to define the end of the frame as well as the beginning of the next frame to distinguish

Approaches of Framing

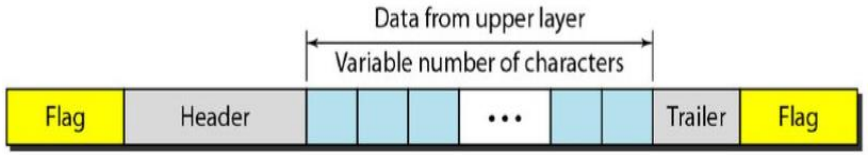
There are two main types of framing approaches used in computer networks: byte-oriented framing and bit-oriented framing.

Bit-oriented Framing

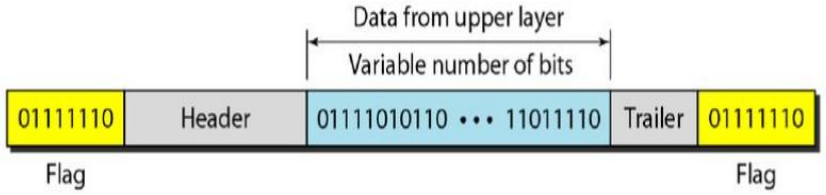
- In **bit-oriented framing**, data is transmitted as a continuous stream of bits, with special **bit patterns** used to indicate the start and end of a frame.
- Uses flag bits (e.g., 01111110 in HDLC).

- Uses bit stuffing to ensure flag pattern does not appear in the payload

Framing in byte-oriented protocols



Framing in bit-oriented protocols



Byte-Oriented Framing (Character-Oriented Framing)

- In **byte-oriented framing**, data is sent as a series of bytes, and special **characters (bytes)** are used to mark the beginning and end of frames.
- Uses special byte values for framing (e.g., SOH, EOT, DLE).
- Byte stuffing is used if control characters appear in the data.
- Control bytes like SOH (Start of Header) or EOT (End of Transmission) are added.

Feature	Bit-Oriented Framing	Byte-Oriented Framing
Framing Marker	Bit pattern (e.g., 01111110)	Special bytes (e.g., SOH, EOT)
Data Handling	Continuous bit stream	Stream of bytes/characters
Stuffing Method	Bit stuffing	Byte stuffing
Efficient for	Binary data	Textual data
Complexity	More complex due to bit-level operations	Simpler for character data
Protocol Example	HDLC, PPP (bit mode)	BISYNC, PPP (byte mode)

- Flow control is a technique that allows two stations working at different speeds to communicate with each other.
- It is a set of measures taken to regulate the amount of data that a sender sends so that a fast sender does not overwhelm a slow receiver.
- In data link layer, flow control restricts the number of frames the sender can send before it waits for an acknowledgment from the receiver.
- Data Link Control is the service provided by the Data Link Layer to provide reliable data transfer over the physical medium.
- For example, In the half-duplex transmission mode, one device can only transmit the data at a time.
- If both the devices at the end of the links transmit the data simultaneously, they will collide and leads to the loss of the information.
- The Data link layer provides the coordination among the devices so that no collision occurs.

- The Data link layer provides following functions.

➢ Flow Control

➢ Error Control

Flow Control:

It is a set of procedures that tells the sender how much data it can transmit before the data overwhelms the receiver.

The receiving device has limited speed and limited memory to store the data.

Therefore, the receiving device must be able to inform the sending device to stop the transmission temporarily before the limits are reached.

It requires a buffer, a block of memory for storing the information until they are processed.

Two methods have been developed to control the flow of data:

- Stop-and-wait
- Sliding window

- **Stop-and-wait:**

Stop and wait means, whatever the data that sender wants to send, he sends the data to the receiver.

After sending the data, he stops and waits until he receives the acknowledgment from the receiver.

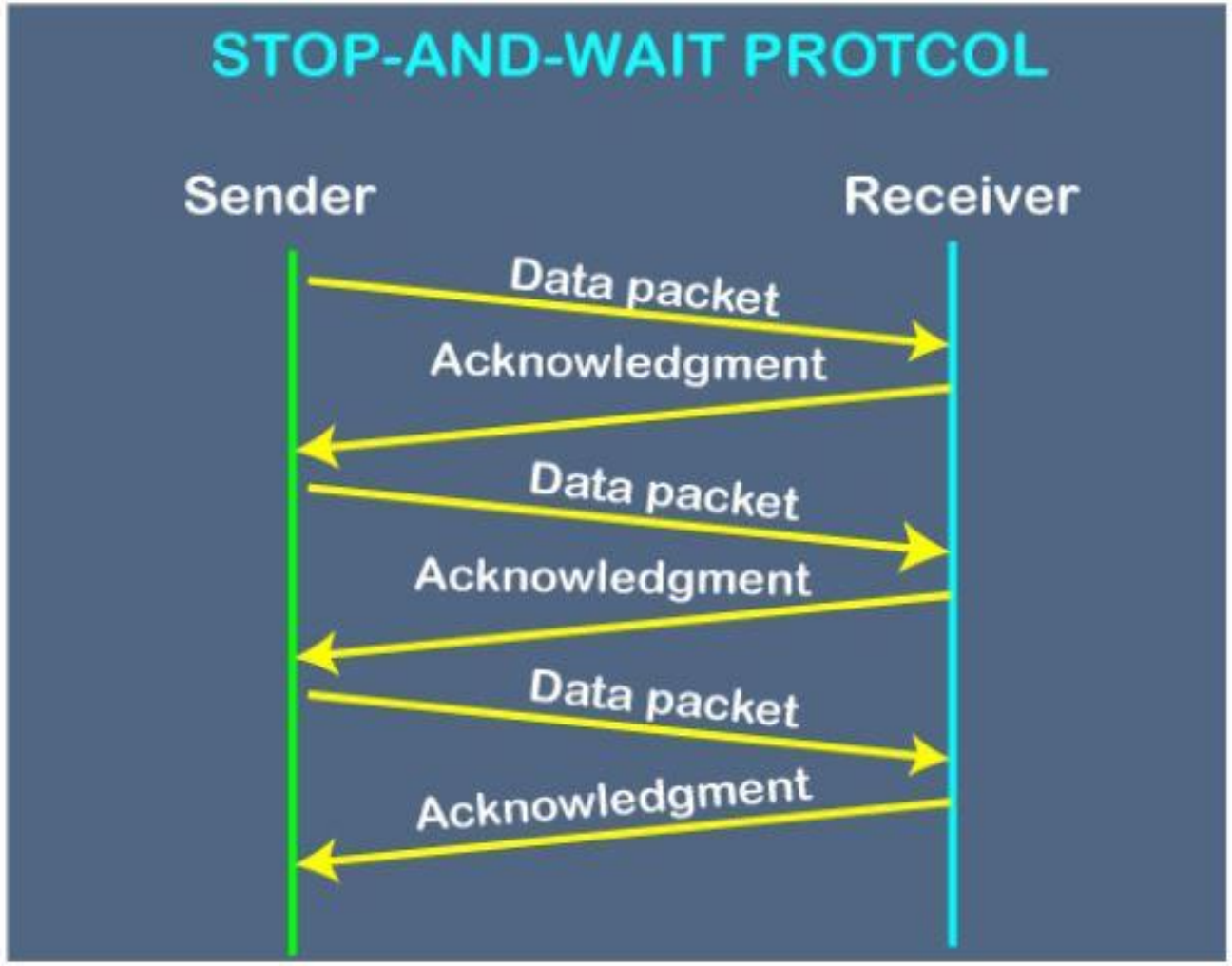
The stop and wait protocol is a flow control protocol where flow control is one of the services of the data link layer.

It is a data-link layer protocol which is used for transmitting the data over the noiseless channels.

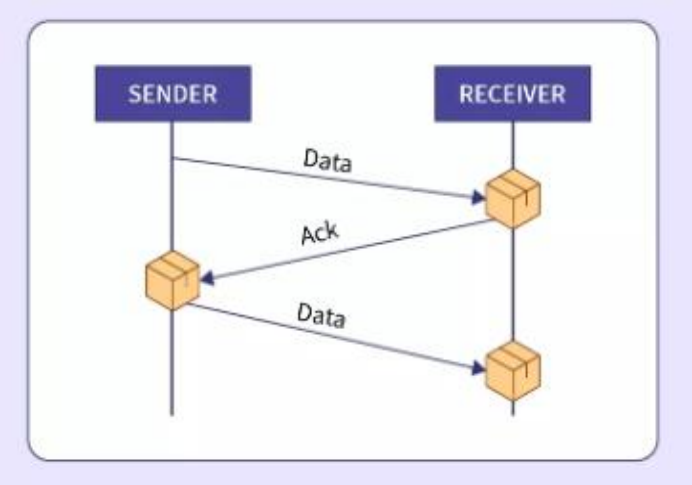
It provides unidirectional data transmission which means that either sending or receiving of data will take place at a time.

It provides flow-control mechanism but does not provide any error control mechanism.

- The idea behind the usage of this frame is that when the sender sends the frame then he waits for the acknowledgment before sending the next frame.



Working of Stop and Wait Protocol



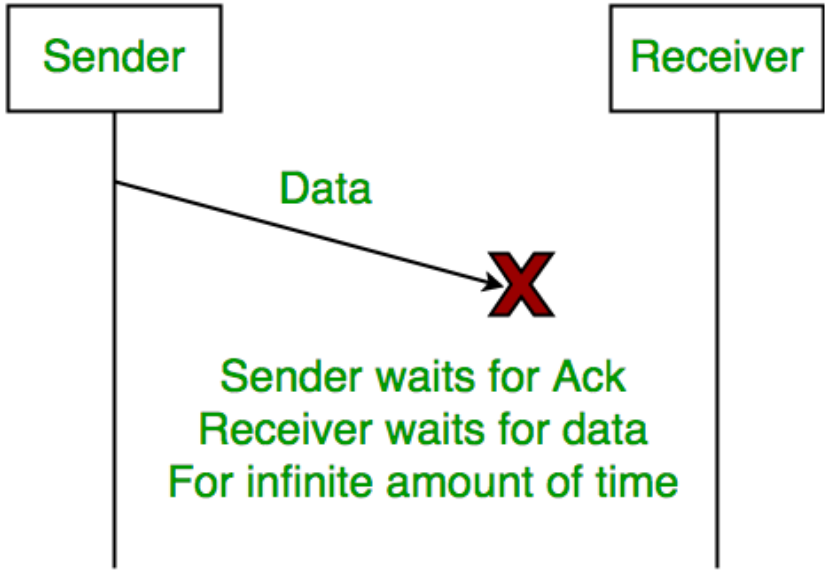
- If there is a sender and receiver, then sender sends the frame and that frame is known as a data packet. The sender will not send the second frame without receiving the acknowledgment of the first frame.
- The receiver sends the acknowledgment for the data frame that it has received. Once the acknowledgment is received, the sender sends the next frame. This process continues until all the frames are not sent.

- The main **advantage** of this protocol is its simplicity but it has some disadvantages also.
- For example, if there are 1000 data frames to be sent, then all the 1000 frames cannot be sent at a time as in Stop and Wait protocol, one frame is sent at a time.

- **Disadvantages of Stop and Wait protocol :**

The following are the problems associated with a stop and wait protocol:

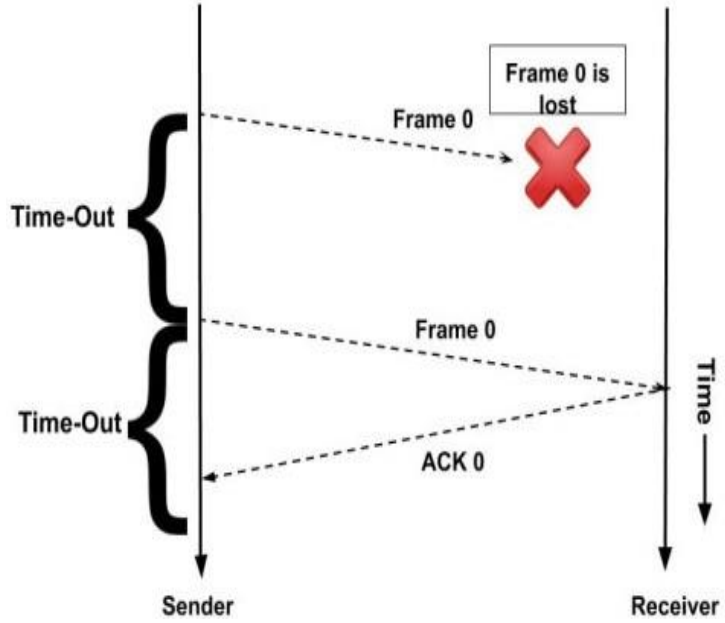
1. **Problems occur due to lost data** Sender waits for an infinite amount of time for an acknowledgment. Receiver waits for an infinite amount of time for a data.

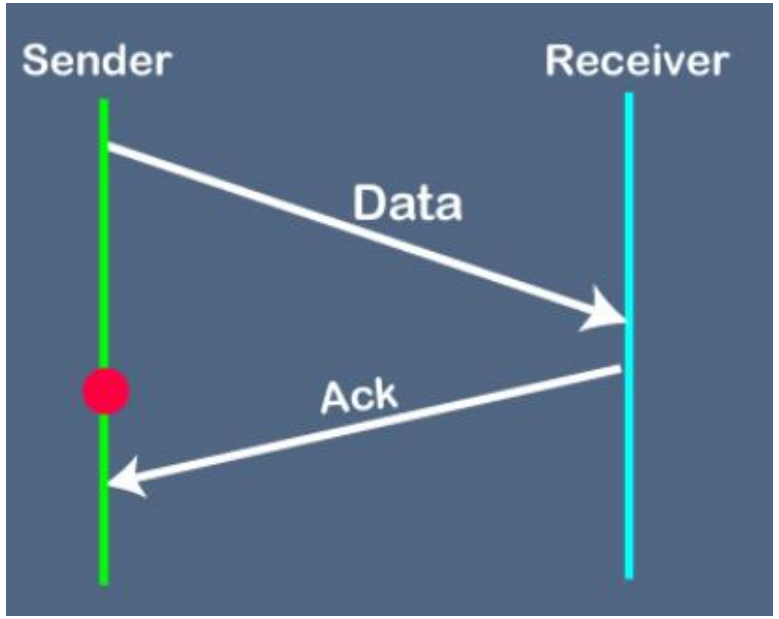


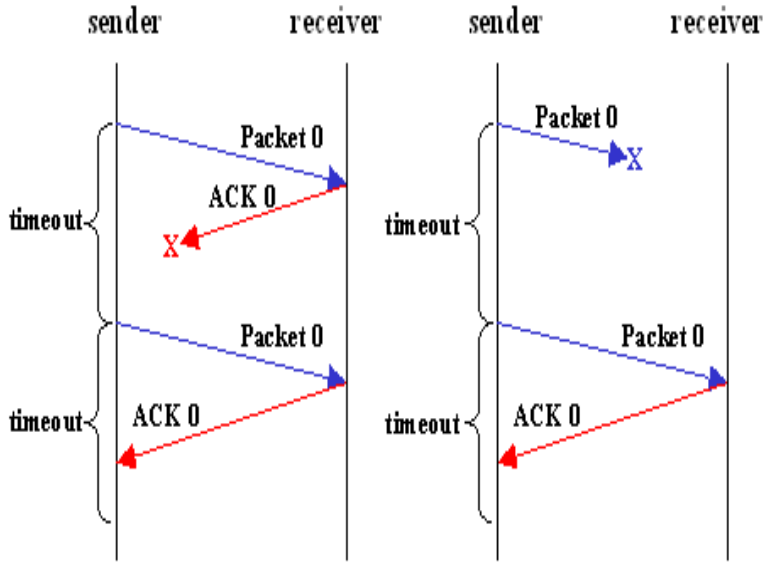
2. **Problems occur due to lost acknowledgment** Sender waits for an infinite amount of time for an acknowledgment.

3. **Problem due to the delayed data or acknowledgment** Suppose the sender sends the data and it has also been received by the receiver. The receiver then sends the acknowledgment but the acknowledgment is received after the timeout period on the sender's side.

As the acknowledgment is received late, so acknowledgment can be wrongly considered as the acknowledgment of some other data frame.







Sliding Window

- The Sliding Window is a method of flow control in which a sender can transmit the several frames before getting an acknowledgement.
- In Sliding Window Control, multiple frames can be sent one after the another due to which **capacity of the communication channel** can be utilized efficiently.
- A single ACK acknowledge multiple frames.
- Sliding Window refers to imaginary boxes at both the sender and receiver end.
- The window can hold the frames at either end, and it provides the upper limit on the number of frames that can be transmitted before the acknowledgement.
- Frames can be acknowledged even when the window is not completely filled.

- The window has a specific size in which they are numbered as modulo-n means that they are numbered from 0 to n-1.
- For example, if n=8, the frames are numbered from 0,1,2,3,4,5,6,7,0,1,2,3,4,5,6,7,0,1..... The size of the window is represented as n-1. Therefore, maximum n-1 frames can be sent before acknowledgement.
- When the receiver sends the ACK, it includes the number of the next frame that it wants to receive.
- For example, to acknowledge the string of frames ending with frame number 4, the receiver will send the ACK containing the number 5.
- When the sender sees the ACK with the number 5, it got to know that the frames from 0 through 4 have been received.

2. Error Control

- Error Control is a technique of error detection and retransmission.
- Error control in data link layer is the process of detecting and correcting data frames that have been corrupted or lost during transmission.
- Data link layer follows a technique to detect transit errors and take necessary actions, which is retransmission of frames whenever error is detected or frame is lost.
- The process is called **Automatic Repeat Request (ARQ).**

Phases in Error Control:

- The error control mechanism in data link layer involves the following phases **Detection of Error** : Transmission error, if any, is detected by either the sender or the receiver.
- **Acknowledgment** – acknowledgment may be positive or negative.
- **Positive ACK** – On receiving a correct frame, the receiver sends a positive acknowledge.
- **Negative ACK** – On receiving a damaged frame or a duplicate frame, the receiver sends a negative acknowledgment back to the sender.
- **Retransmission** – The sender maintains a clock and sets a timeout period. If an acknowledgment of a data-frame previously transmitted does not arrive before the timeout, or a negative acknowledgment is received, the sender retransmits the frame.

Working Principle of ARQ

- The main function of these protocols is, the sender receives an acknowledgement from the receiver end implying that the frame or packet is received correctly before a timeout occurs, timeout is a specific time period within which the acknowledgement has to be sent by the receiver to the sender.
- **If a timeout occurs**: the sender does not receive the acknowledgement before the specified time, it is implied that the frame or packet has been corrupt or lost during the transmission.
- Accordingly, the sender retransmits the packet and these protocols ensure that this process is repeated until the correct packet is transmitted.

Error Control
Stop-and-wait ARQ
Sliding window ARQ
Go-back-n
Selective-reject

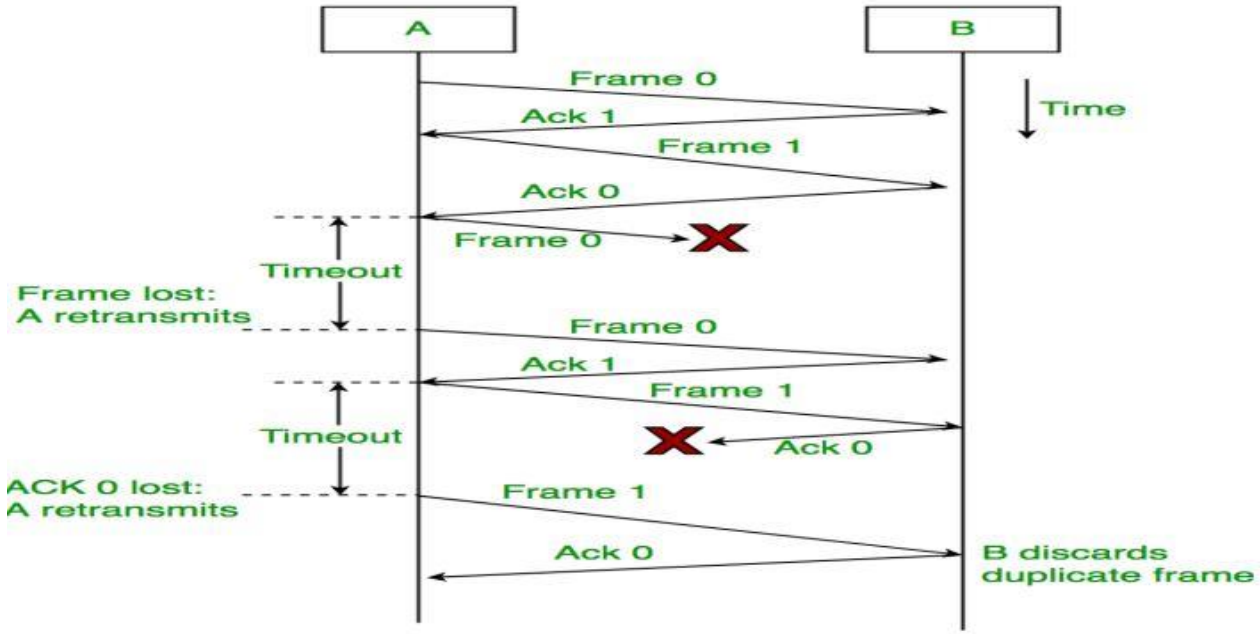
Stop-and-wait ARQ

- Stop-and-wait ARQ is a technique used to retransmit the data in case of damaged or lost frames.
- This technique works on the principle that the sender will not transmit the next frame until it receives the acknowledgement of the last transmitted frame.

Four features are required for the retransmission:

- The sending device keeps a copy of the last transmitted frame until the acknowledgement is received.
- Keeping the copy allows the sender to retransmit the data if the frame is not received correctly.
- Both the data frames and the ACK frames are numbered alternately 0 and 1 so that they can be identified individually.
- Suppose data 1 frame acknowledges the data 0 frame means that the data 0 frame has been arrived correctly and expects to receive data 1 frame.

- If an error occurs in the last transmitted frame, then the receiver sends the NAK frame which is not numbered.
- On receiving the NAK frame, sender retransmits the data.
- It works with the timer. If the acknowledgement is not received within the allotted time, then the sender assumes that the frame is lost during the transmission, so it will retransmit the frame.



Two possibilities of the retransmission:

Damaged Frame: When the receiver receives a damaged frame, i.e., the frame contains an error, then it returns the NAK frame.

For example, when the data 0 frame is sent, and then the receiver sends the ACK 1 frame means that the data 0 has arrived correctly, and transmits the data 1 frame.

The sender transmits the next frame: data 1. It reaches undamaged, and the receiver returns ACK 0.

The sender transmits the next frame: data 0. The receiver reports an error and returns the NAK frame. The sender retransmits the data 0 frame.

Lost Frame: Sender is equipped with the timer and starts when the frame is transmitted.

Sometimes the frame has not arrived at the receiving end so that it can be acknowledged neither positively nor negatively.

The sender waits for acknowledgement until the timer goes off.

If the timer goes off, it retransmits the last transmitted frame

Sliding Window ARQ

- Sliding Window ARQ is a technique used for continuous transmission error control.

Three Features used for retransmission:

- In this case, the sender keeps the copies of all the transmitted frames until they have been acknowledged.
- Suppose the frames from 0 through 4 have been transmitted, and the last acknowledgement was for frame 2, the sender has to keep the copies of frames 3 and 4 until they receive correctly.

- The receiver can send either NAK or ACK depending on the conditions.
- The NAK frame tells the sender that the data have been received damaged.
- Since the sliding window is a continuous transmission mechanism, both ACK and NAK must be numbered for the identification of a frame.
- The ACK frame consists of a number that represents the next frame which the receiver expects to receive.
- The NAK frame consists of a number that represents the damaged frame.

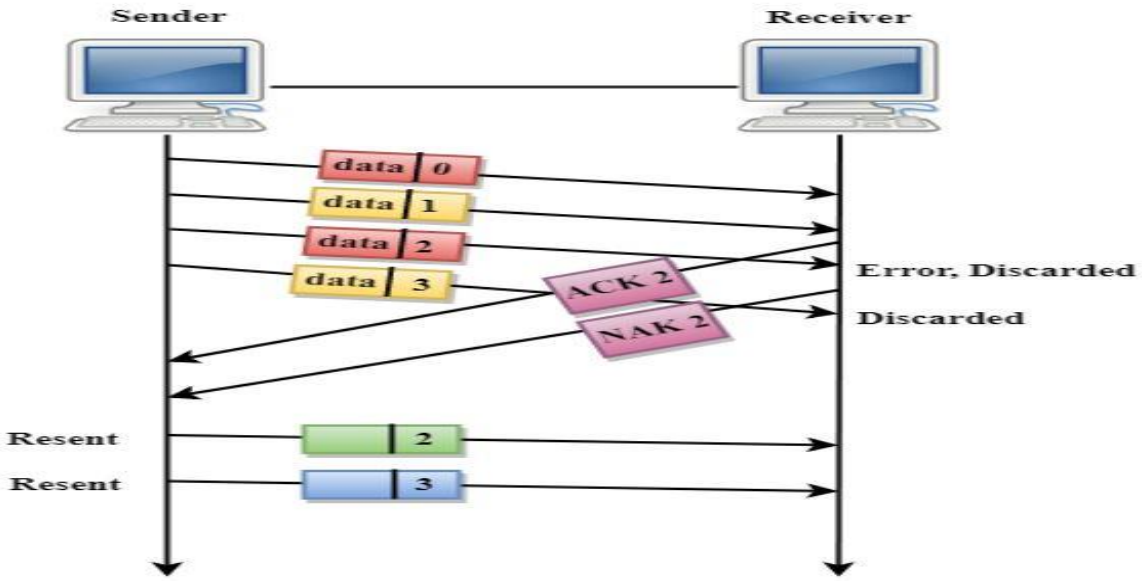
The sliding window ARQ is equipped with the timer to handle the lost acknowledgements.

- Suppose then n-1 frames have been sent before receiving any acknowledgement.
- The sender waits for the acknowledgement, so it starts the timer and waits before sending any more.
- If the allotted time runs out, the sender retransmits one or all the frames depending upon the protocol used.

- Two protocols used in sliding window ARQ:
- **Go-Back-n ARQ:**
- In Go-Back-N ARQ protocol, if one frame is lost or damaged, then it retransmits all the frames after which it does not receive the positive ACK.

Three possibilities can occur for retransmission:

- **Damaged Frame**: When the frame is damaged, then the receiver sends a NAK frame.



- In the above figure, three frames have been transmitted before an error discovered in the third frame.
- In this case, ACK 2 has been returned telling that the frames 0,1 have been received successfully without any error.
- The receiver discovers the error in data 2 frame, so it returns the NAK 2 frame.
- The frame 3 is also discarded as it is transmitted after the damaged frame. Therefore, the sender retransmits the frames 2,3.

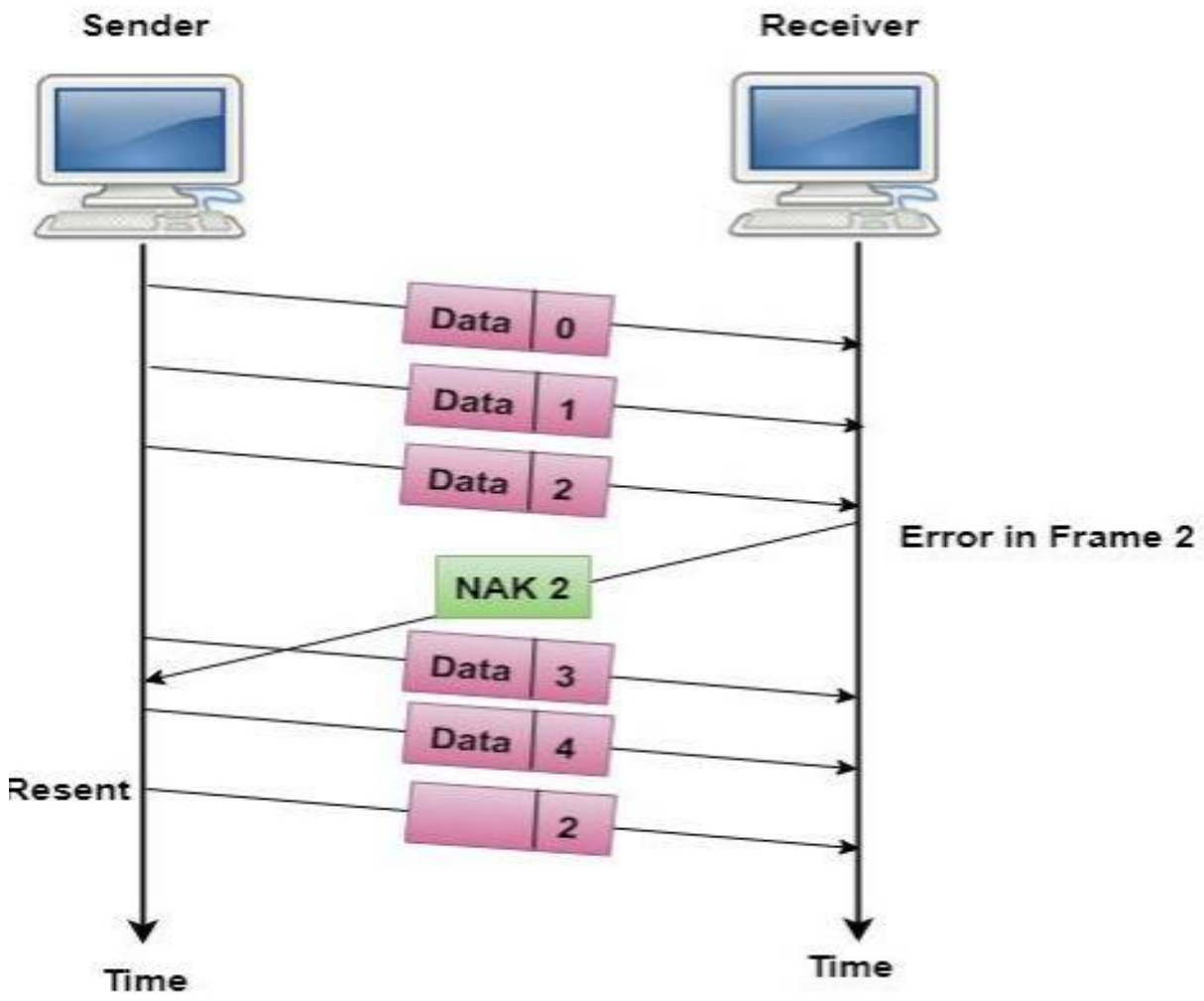
- **Lost Data Frame**: In Sliding window protocols, data frames are sent sequentially. If any of the frames is lost, then the next frame arrive at the receiver is out of sequence.
- The receiver checks the sequence number of each of the frame, discovers the frame that has been skipped, and returns the NAK for the missing frame.
- The sending device retransmits the frame indicated by NAK as well as the frames transmitted after the lost frame.
- **Lost Acknowledgement**: The sender can send as many frames as the windows allow before waiting for any acknowledgement.

- Once the limit of the window is reached, the sender has no more frames to send; it must wait for the acknowledgement.
- If the acknowledgement is lost, then the sender could wait forever.
- To avoid such situation, the sender is equipped with the timer that starts counting whenever the window capacity is reached.
- If the acknowledgement has not been received within the time limit, then the sender retransmits the frame since the last ACK.

2.Selective-Repeat ARQ

- Selective-Repeat ARQ technique is more efficient than Go-Back-n ARQ.
- In this technique, only those frames are retransmitted for which negative acknowledgement (NAK) has been received.
- The receiver storage buffer keeps all the damaged frames on hold until the frame in error is correctly received.
- The receiver must have an appropriate logic for reinserting the frames in a correct order.
- The sender must consist of a searching mechanism that selects only the requested frame for retransmission.
- The Go-back-N ARQ protocol works well if it has fewer errors. But if there is a lot of error in the frame, **lots of bandwidth loss in sending the frames again**.

- So, we use the Selective Repeat ARQ protocol.
- In this protocol, the size of the sender window is always equal to the size of the receiver window.
- The size of the sliding window is always greater than 1. If the receiver receives a corrupt frame, it does not directly discard it.
- It sends a negative acknowledgment to the sender.
- The sender sends that frame again as soon as on the receiving negative acknowledgment.
- There is no waiting for any time-out to send that frame.


Sender
Receiver
Data 0
Data 1
Data 2
Error in Frame 2
NAK 2
Data 3
Data 4
Resent
2
Time
Time

Steps for the Sender Side

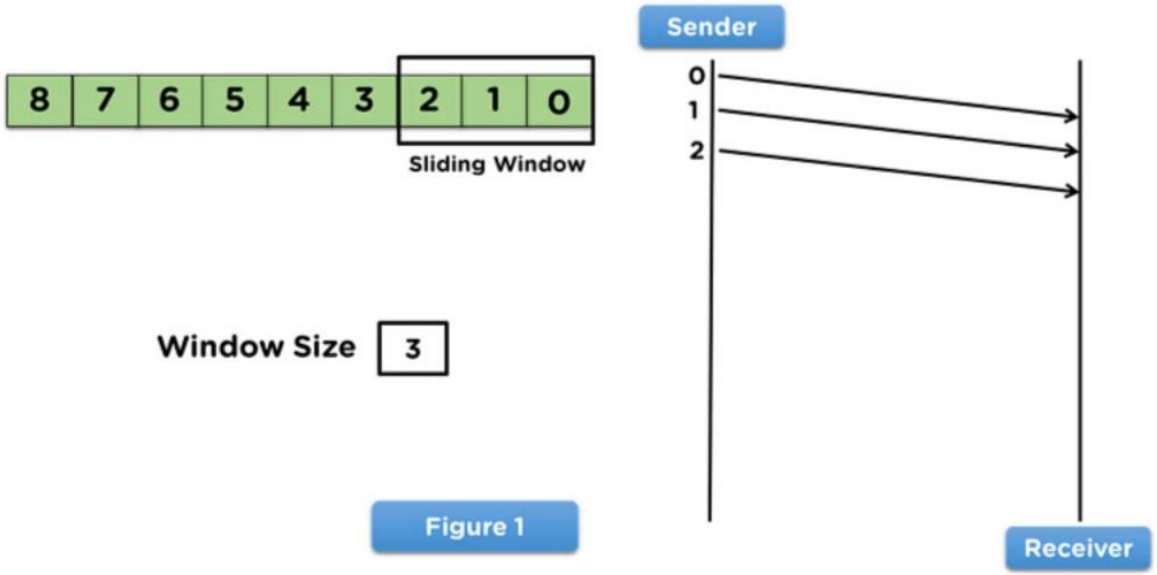


Figure 1

Steps for the Receiver Side

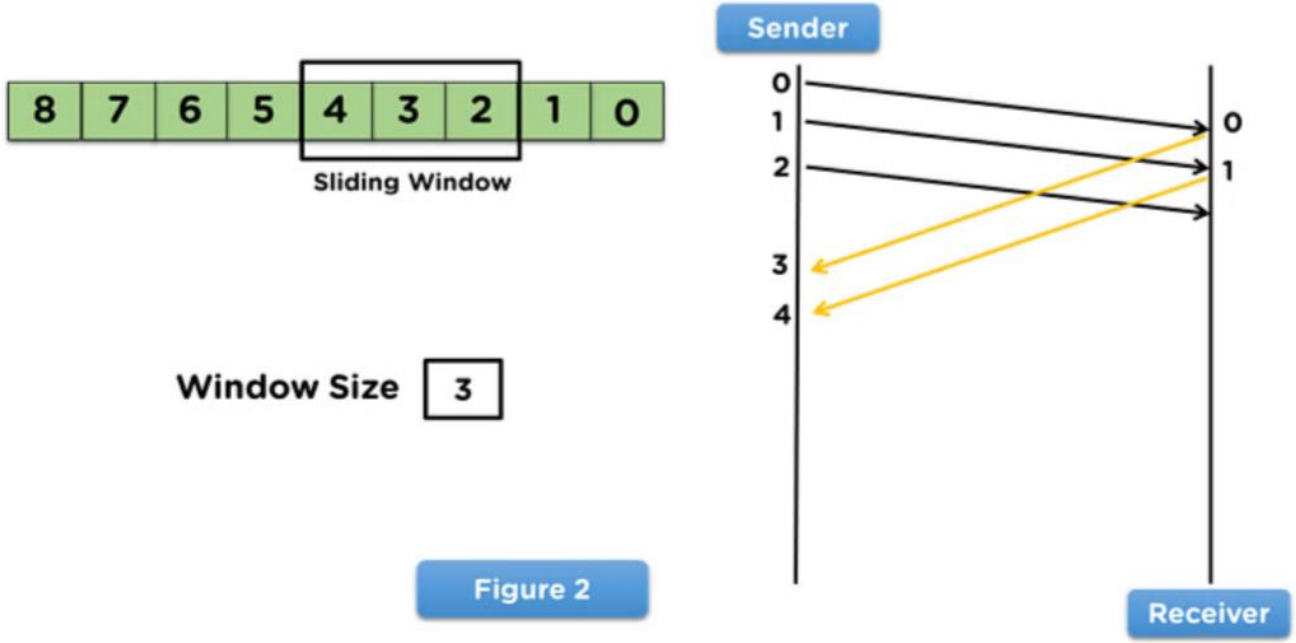


Figure 2

High-Level Data Link Control (HDLC)

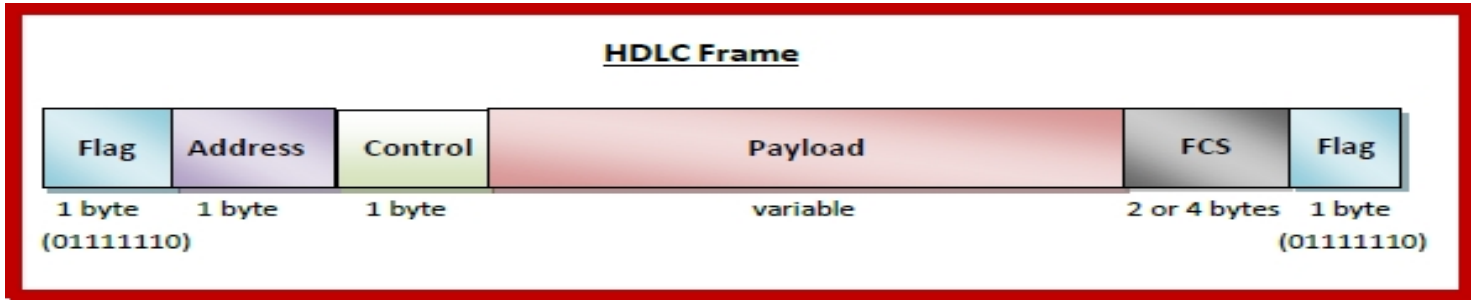
- High-level Data Link Control (HDLC) is a group of communication protocols of the **data link layer** for transmitting data between network points or nodes. Since it is a **data link protocol**, data is organized into frames.
- A frame is transmitted via the network to the destination that verifies its successful arrival. It is a bit - oriented protocol that is applicable for both point - to - point and multipoint communications.

Transfer Modes

- HDLC supports two types of transfer modes, normal response mode and asynchronous balanced mode.

HDLC Frame

- HDLC is a bit - oriented protocol where each frame contains up to **six fields**. The structure varies according to the type of frame. The fields of a HDLC frame are –



Flag – It is an **8-bit sequence** that marks the beginning and the end of the frame. The bit pattern of the flag is **01111110.**

Address – It contains the **address of the receiver**. If the frame is sent by the **primary station**, it contains the **address(es) of the secondary station(s)**. If it is sent by the **secondary station**, it contains the address of the **primary station**. The address field may be from **1 byte to several bytes.**

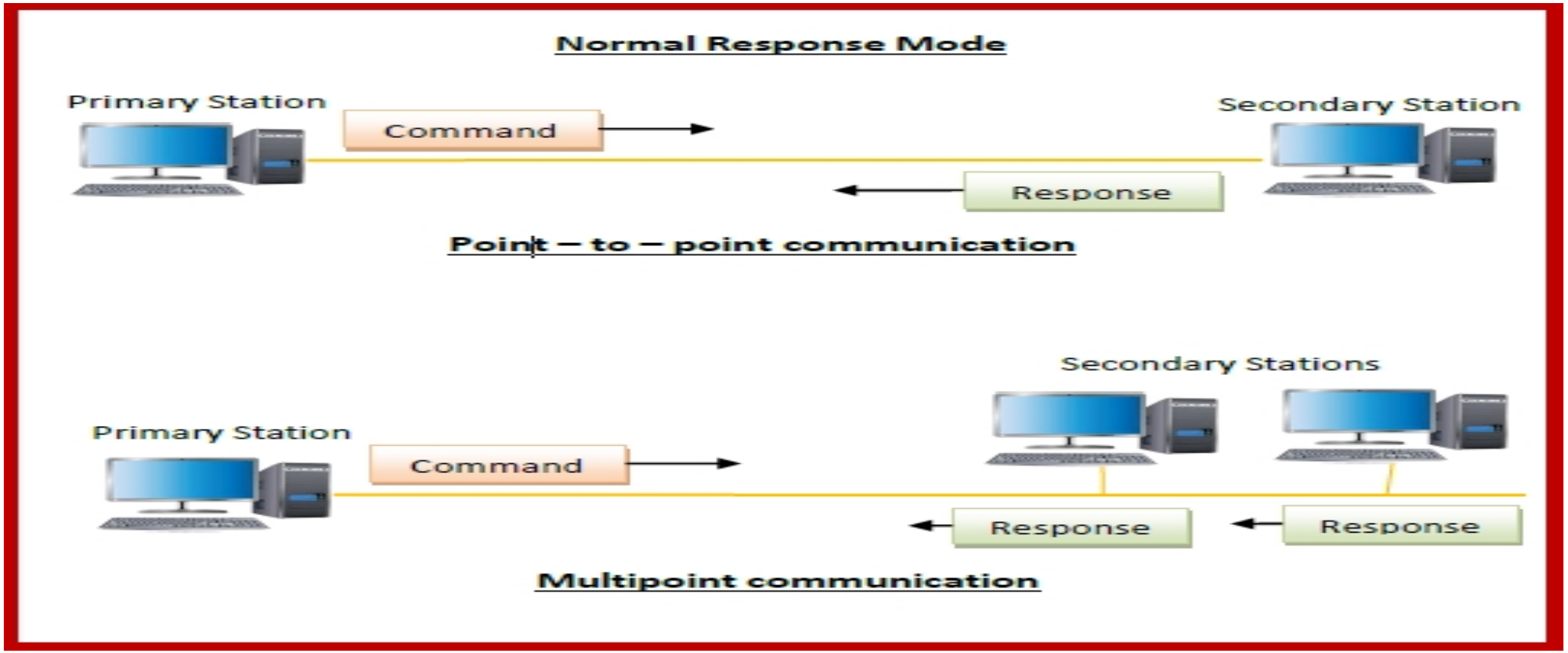
Control – It is **1 or 2 bytes** containing **flow and error control** information.

Payload – This carries the data from the network layer. Its **length may vary** from one network to another.

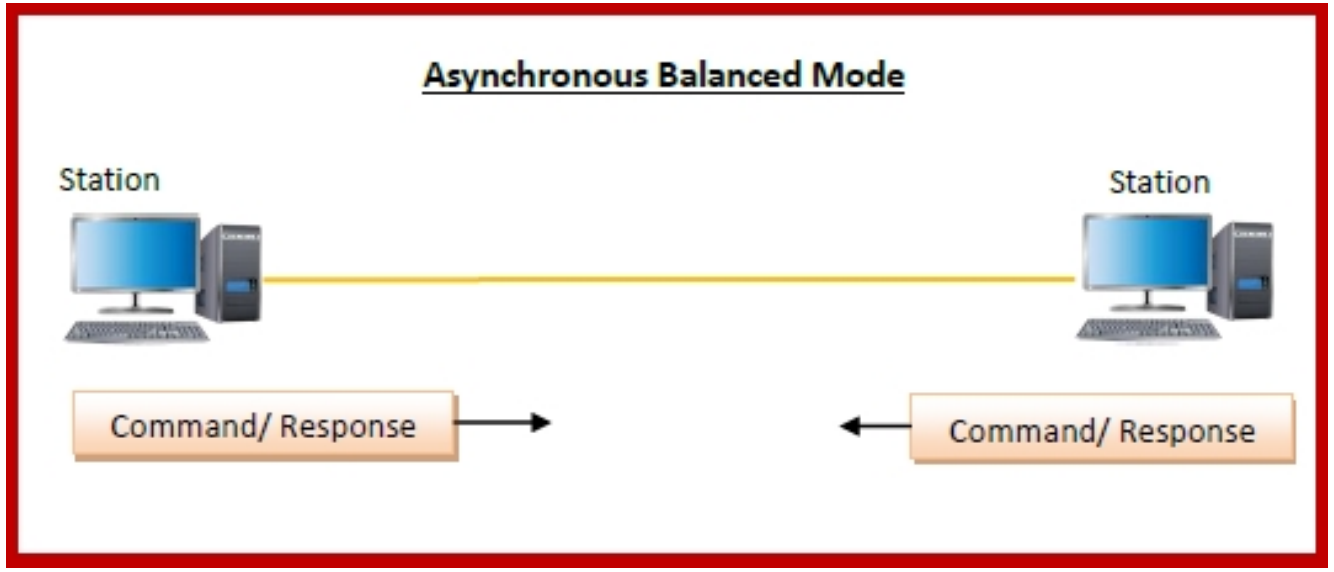
FCS – It is a **2 byte or 4 bytes frame check sequence** for **error detection**. The standard code used is CRC (cyclic redundancy code)

Normal Response Mode (NRM) :

Here, two types of stations are there, a primary station that send commands and secondary station that can respond to received commands. It is used for both point - to - point and multipoint communications.

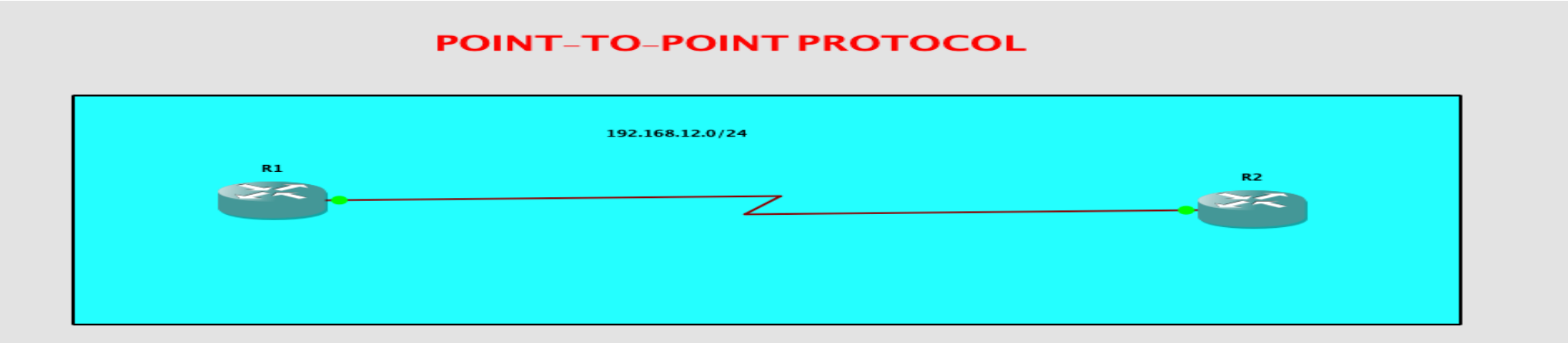


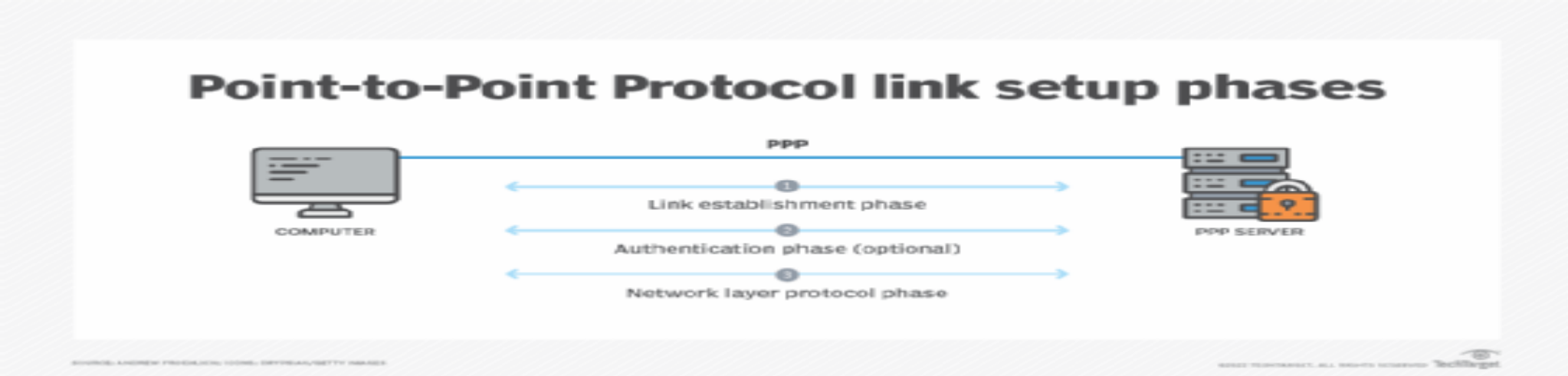
- **Asynchronous Balanced Mode (ABM)** – Here, the configuration is balanced, i.e. each station can both send commands and respond to commands. It is used for only point - to - point communications.



Point-to-Point Protocol (PPP)

- Point-to-Point Protocol (PPP) is a data link layer communication protocol that enables the direct transmission of data between two network nodes. It is commonly used to establish connections over various physical mediums, such as serial cables, phone lines, and fiber optics, making it a versatile protocol for network communication. PPP provides features like error detection, authentication, and data compression, enhancing the reliability and security of data transfer.





Components of PPP

Point - to - Point Protocol is a layered protocol having three components

- **Encapsulation Component:** It encapsulates **network layer protocols** so that it can be transmitted over the specified physical layer.
- **Link Control Protocol (LCP):** It is responsible for **establishing, configuring, testing, maintaining and terminating links** for transmission. It also imparts negotiation for set up of options and use of features by the two endpoints of the links.
- **Authentication Protocols (AP):** These protocols **authenticate endpoints** for use of services. The two authentication protocols of PPP are

➢ **Password Authentication Protocol** (PAP)

➢ **Challenge Handshake Authentication Protocol** (CHAP)

- **Network Control Protocols (NCPs):**

NCPs are used to establish and configure specific network layer protocols, ensuring proper communication between the two nodes.

PPP Frame

PPP is a **byte - oriented protocol** where each field of the frame is composed of one or more bytes. The fields of a PPP frame are

•PPP frame similar to HDLC frame, always begins and ends with standard HDLC flag. **Flag** 1 byte that marks the beginning and the end of the frame. The bit pattern of the flag is 01111110.

•**Address: Address field** is basically **broadcast address**. In this, all 1's simply indicates that all of the stations are ready to accept frame.1 byte which is set to 11111111 in case of broadcast.

•**Control field: In** HDLC, control field is required for various purposes but in PPP, this **field is set to 1 byte** i.e., 00000011 binary value. This 1 byte is used for a connection-less data link.

•**Protocol: This** field basically identifies network protocol of the datagram. It usually identifies the kind of packet in the data field .1 or 2 bytes that define the type of data contained in the payload field.

•**Payload** This carries the data from the network layer. **The maximum length of the payload field is 1500 bytes**. However, this may be negotiated between the endpoints of communication.

•**FCS(Frame check sequence):** It is a **2 byte or 4 bytes** frame check sequence **for error detection**. The standard code used is CRC (cyclic redundancy code)

END OF CHAPTER-7

THANK YOU